

Rohit Suryaa Saravanan

Software Developer and Data Scientist

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SUMMARY

Software developer and data scientist, currently doing an MSc in Data Science at Hochschule Fulda. One year of professional experience building enterprise software for a US cybersecurity startup, two IEEE papers in applied machine learning, and practical work on AI agents built with LangGraph and RAG. I care about the model, and just as much about the API around it, the pipeline that ships it, and whether the result actually helps.

EXPERIENCE

Associate Software Developer · EigenSecure, remote (USA)

Mar 2024 to Apr 2025

- Turned business and security requirements into backend features, REST APIs and proof of concept components in an Agile Scrum team.
- Built features around access control, evidence workflows and compliance data, working across MySQL, PostgreSQL and MongoDB.
- Improved SQL queries, data workflows and CI/CD pipelines for reliable deployments and traceable audit trails.
- Kept engineering and product stakeholders informed about progress and the choices involved, balancing security needs with delivery speed.

SELECTED PROJECTS

JUMO Laser Trimming · Industrial ML, MSc team project with JUMO

Apr to Jul 2026

- Predicted the laser correction for platinum temperature sensors from eight production files (47,115 substrates). Owned the framing, source inventory, join audits, part number parser, trust controls, the NCTr target fix and the end to end smoke test.
- Optuna tuned XGBoost cut RMSE from 0.1044 (linear baseline) to 0.0202 with 99.63% inside the ± 0.1 tolerance; tolerance class model at 0.944 weighted F1. Served through FastAPI and React with MLflow model versions, AI labels and an audit history, deployed to Azure.

Stack: Python, XGBoost, Optuna, MLflow, FastAPI, React, SQLite, Azure · Team of six, Hochschule Fulda

Windmill Pitch Prediction · IEEE ICEMCE 2023

Feb to Jun 2024

- Published version: 96% accuracy, error cut from 8.6% to 2.3% and modelled output up 21% (841.54 to 1,019.41 kWh on a sample day).
- Rebuilt on six months of simulated 10 minute data for three 250 kW turbines: 26 features, XGBoost plus a second model trained on its errors, pitch error down from 33.95% (linear baseline) to 1.11%.

Stack: Python, XGBoost, time series, simulation, statistical evaluation · DOI 10.1109/ICEMCE57940.2023.10433983

Supply Chain Intelligence · Forecasting and analytics pipeline

2025

- Forecast weekly demand on 110,123 real Olist orders. With 86 weeks of history, SARIMA beat XGBoost and every baseline (MAPE 18.5% against 26.9%), inside the 20% benchmark.
- Added EOQ and safety stock, delivery analysis (promised 23.4 days, delivered in 11.9, 92.1% on time) and risk scores for 881 sellers, in a Plotly Dash dashboard.

Stack: Python, Pandas, NumPy, SARIMA, XGBoost, SQL, Plotly Dash · github.com/srohitsuryaa21/supply-chain-forecasting

Luxury Concierge Agent · AI agent for vehicle configuration

2026

- Built a five step LangGraph agent over a SQLite catalogue that turns usage, region, cabin style, budget and timeline into a priced specification ready for sales.
- Every answer is checked against the catalogue with Pydantic, budgets are enforced in Python so the model can never set a price, conversations are saved, and FastAPI serves a hand written front end. Twenty offline test scenarios run without a single API call.

Stack: Python, LangGraph, RAG, FastAPI, Pydantic, SQLite, pytest · github.com/srohitsuryaa21/luxury-concierge-agent

FinGuard PayOps · Payment operations platform

- Payment operations platform: seven explainable fraud rules add risk points (review from 70, block from 90), with every reason stored, compliance alerts, an analyst approve or reject flow and an audit log of every step.
- Java 21 and Spring Boot backend, React dashboard and a Python scoring service, with tests for the fraud rules and incident workflow.

Stack: Java, Spring Boot, PostgreSQL, React · github.com/srohitsuryaa21/FinGuard-PayOps

▸ Wrote Ethereum smart contracts for loans, payments, insurance and supply chain tracking for small farmers, with a Web3.py backend tested on a local Ganache network. Modelled adoption of digital records rose from under 50% to about 80%.
Stack: Solidity, Ethereum, Web3.py, Ganache · DOI 10.1109/ICSCSS60660.2024.10625462

Aviation Delay Ontology · Data pipeline and Power BI

▸ Modelled 63,029 US flights from June 2018 as connected flight and airport objects in the style of Palantir Foundry, with the same transforms in pandas and PySpark, reported in Power BI.
▸ Chicago O'Hare averaged 21.1 minutes of arrival delay against 13.6 at Atlanta, with 32% of flights delayed against 24%.
Stack: Python, pandas, PySpark, Power BI · github.com/srohitsuryaa21/aviation-delay-ontology

SOC Silverstone · Formula 1 telemetry analysis

2026

▸ Inferred battery deployment from speed and throttle in public FastF1 telemetry for the 2026 British Grand Prix: the speed stalled while flat out on 76% of 5,540 runs down a straight.
▸ Across 90 tyre stints, deployment and tyre wear were unrelated ($r = -0.04$); per session results pointed in opposite directions, a small sample trap. Deployment mattered for straight line speed: Antonelli gained 7.9 km/h over his teammate.
Stack: Python, FastF1, statistics · github.com/srohitsuryaa21/SOC-Silverstone-2026

Employee Attrition Prediction · Explainable machine learning

Dec 2022 to Feb 2023

▸ On IBM's HR dataset (1,470 employees, 16.1% left), compared four models by AUC rather than accuracy; logistic regression won with AUC 0.80. Found and replaced a synthetic data stand-in that had capped results near chance.
▸ Explained the drivers: 30.5% of employees on overtime left against 10.4% without; sales representatives left at 39.8%.
Stack: Python, Scikit-learn, Pandas, classification, explainability · github.com/srohitsuryaa21/employee-attrition-prediction

Personal portfolio · rohitsuryaa.com

2025 to now

▸ Bilingual website in English and German where every case study is an interactive scroll story built from the project's own data, with full fallbacks for reduced motion and screen readers.
Stack: Astro, TypeScript, GSAP, CSS · rohitsuryaa.com

EDUCATION

MSc Data Science · Hochschule Fulda, Germany

Apr 2025 to now

Machine learning, big data and distributed systems, data mining and predictive analytics, process mining, visualisation and BI, research methods, data ethics and GDPR.

BTech Big Data Computer Science · SRM University, Chennai, India

Aug 2020 to Nov 2024

Big data systems, distributed systems, data structures and algorithms, databases, software engineering.

TECHNICAL SKILLS

Data and ML	Python, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, TensorFlow, SARIMA, time series, classification, explainability, statistics
AI agents	LangGraph, RAG, FastAPI, Pydantic, ChromaDB, Ollama, pytest, offline evaluation
Software	C#, .NET, ASP.NET Core, Blazor, Java, Spring Boot, REST APIs, JavaScript, TypeScript, React
Data and BI	SQL, PostgreSQL, MySQL, MongoDB, SQLite, PySpark, ETL, Power BI, Plotly Dash, Tableau
Cloud and delivery	Azure, Docker, GitHub Actions, Jenkins, CI/CD, Git, Linux

PUBLICATIONS

▸ Windmill pitch prediction with XGBoost and residual correction. IEEE ICEMCE, 2023. DOI 10.1109/ICEMCE57940.2023.10433983
▸ A DeFi platform for transparent agricultural finance. IEEE ICSCSS, 2024. DOI 10.1109/ICSCSS60660.2024.10625462

CERTIFICATIONS

LANGUAGES

IBM Data Science Professional (Coursera)

English: fluent (C1+) · German: A2, improving · Tamil: native